

PRABHAT DHAR

Data Architect | Cloud Data Engineer | Machine Learning Specialist

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🐙 github.com/Xclipxz07 🌐 xclipxz07.github.io/Portfolio 🇬🇧 **UK Work Status:** Eligible (20 hrs/wk)

PROFESSIONAL SUMMARY

Physics-grounded **Data Architect & Machine Learning Engineer** with a B.Sc. in Physics and a Postgraduate Diploma in Data Science from the University of Essex. Proven expertise engineering end-to-end cloud lakehouses, sub-millisecond AI surrogate models, and automated CI/CD pipelines. Specializes in building high-throughput, cost-optimized AWS architectures (S3, Glue, Athena, EMR PySpark), Apache Airflow orchestration, Snappy Parquet partitioning, and production MLOps deployments.

EDUCATION & ACADEMIC BACKGROUND

Postgraduate Diploma in Data Science & Its Applications

University of Essex, Colchester, United Kingdom
November 2022 • 165 Credits (Distinction Level)

I.S.C (12th Grade - Pure Science & Mathematics)

St. Xavier's Institution, Kolkata, India
April 2017 • 85.0%

Bachelor of Science in Physics (Honours)

University of Calcutta, West Bengal, India
August 2021 • 7.240 CGPA

I.C.S.E (10th Grade)

St. Xavier's Institution, Kolkata, India
April 2015 • 93.8%

TECHNICAL SKILLS & ARSENAL

Programming Languages:	Python, Advanced SQL (Window Functions, CTEs, Execution Plans), R, Bash/Shell, JavaScript, LaTeX
Cloud & Lakehouse:	Amazon AWS (S3, Glue, Athena, Redshift, EMR PySpark, KMS, QuickSight), Google Cloud Platform (BigQuery)
Data Engineering:	Apache Airflow, PySpark / Spark, DuckDB, Pandas, NumPy, Snappy Parquet, Power Query, JSON/CSV/XML ETL
Machine Learning & AI:	Scikit-Learn, XGBoost, AWS AutoGluon AutoML, PyTorch, SciPy Optimization, OpenCV, OpenAI GPT-4, Gemini API
DevOps, IaC & CI/CD:	Terraform, Docker Containerization, Kubernetes, Jenkins CI/CD Automation, Git, GitHub Actions
BI & Visualization:	Microsoft Power BI, Microsoft Excel & Copilot AI, Plotly, Streamlit, Seaborn, FastAPI REST APIs

PROFESSIONAL WORK EXPERIENCE

Care Assistant & Data Analyst — EVERYCARE LTD

April 2023 – June 2026 | London, UK

- Delivered person-centred care and maintained digital care observation logs, medication updates, and activity tracking in digital care systems while ensuring strict GDPR and safeguarding compliance.
- Maintained and cleaned operational care records in Excel; engineered automated pivot models and dashboards tracking incident logs, attendance patterns, and staff allocations for shift leaders.
- Analyzed care-related trends using Power BI to visualize medication timings, risk indicators, and response times; communicated weekly operational insights to senior management.
- Wrote basic Python automation scripts to clean and preprocess raw care logs, reducing manual administrative workload by 40% and improving data accuracy for management audits.

Team Member — KFC (Demipower Ltd)

January 2021 – March 2023 | Wood Green, London, UK

- Delivered fast, friendly service in a high-pressure environment; upheld food hygiene, workplace safety, and operational quality standards.
- Handled customer transactions, managed order throughput during peak rush periods, and assisted shift leaders by maintaining inventory tracking and checklists.
- Developed cross-functional teamwork, punctuality, and time-critical communication under pressure — key transferable skills for collaborative data engineering.

FEATURED ENGINEERING & MACHINE LEARNING PROJECTS

-  **Industrial EV Digital Twin & AI Physics Surrogate** | [GitHub: Xclipxz07/ai-mechanical-engineering-digital-twin](#)Python, Scikit-Learn, SciPy, DuckDB, Parquet, Airflow, Terraform
 - Engineered a physics-informed EV digital twin ingesting multi-channel battery thermal and autonomous kinematics telemetry streams.
 - Trained AI surrogate neural regressors predicting 8-hour 3D CFD aerodynamic drag forces in **< 0.05 ms ($R^2 > 0.99$)**.
 - Implemented battery thermal gradient (ΔT) monitoring for thermal runaway prevention and applied **SciPy SLSQP optimization** for active aero flap tuning.
-  **AWS Serverless Data Analytics Lakehouse** | [GitHub: Xclipxz07/aws-serverless-data-analytics-lakehouse](#)Amazon S3, AWS Glue, Amazon Athena, Amazon QuickSight, Terraform, SQL
 - Architected an end-to-end serverless data lakehouse and SLA monitoring pipeline on AWS with automated Glue Data Catalog crawlers.
 - Slashed Athena byte-scan query costs by **85%** using Snappy Parquet partitioning and 10 GB per-query circuit breakers.
 - Executed complete 4-pillar analytical suites: Descriptive, Diagnostic, Predictive, and Prescriptive SQL for executive BI.
-  **Channel 4 Big Data Ad Revenue Pipeline** | [GitHub: Xclipxz07/channel4-bigdata-ad-pipeline](#)PySpark, AWS EMR, Apache Airflow, DuckDB, Snappy Parquet, Terraform
 - Automated nightly 2:00 AM Big Data lakehouse ingesting multi-format ad logs (JSON, CSV, XML) across Smart TVs and Web players.
 - Orchestrated PySpark transformations on AWS EMR with Apache Airflow DAGs, computing regional eCPM yield curves in sub-second queries.
-  **Sales Forecasting System for DevOps, MLOps & GenAI** | [GitHub: Xclipxz07/DATA-ANALYSIS-FOR-DEVOPS-MLOPS-GENAI](#)FastAPI, OpenAI GPT-4, Scikit-learn, Docker, Kubernetes, Jenkins, Power BI
 - Built production-ready sales forecasting system combining ML, generative AI (OpenAI GPT-4), and full-lifecycle DevOps automation.
 - Containerized REST API with Docker, orchestrated on Kubernetes with horizontal scaling, and automated testing via Jenkins CI/CD.
-  **Australian Rainfall Forecasting with PyCaret & ML (Dissertation)** | [GitHub: Xclipxz07/Rainfall-prediction](#)Python, PyCaret AutoML, Scikit-learn, XGBoost, Pandas, Statsmodels, LaTeX
 - Conducted academic research dissertation forecasting weather patterns across Australia; engineered automated EDA and outlier handling.
 - Trained and evaluated XGBoost, Random Forests, and Gradient Boosting models using PyCaret AutoML with complete ROC-AUC diagnostics.
-  **NHS Patient Healthcare Analytics Lakehouse** | [GitHub: Xclipxz07/nhs-patient-healthcare-analytics-lakehouse](#)Terraform, AWS S3, AWS Glue, Amazon Athena, Apache Airflow, Snappy Parquet
 - Developed an automated healthcare lakehouse on AWS using Terraform and Airflow for hospital emergency A&E waiting-time analytics.
 - Enforced strict GDPR / HIPAA compliance by implementing SHA-256 cryptographic hashing to pseudonymize sensitive patient identifiers.

VERIFIED CERTIFICATIONS & CLEARANCES

- English SELT ISE 1:** Trinity College London ([Distinction in Speaking, Listening & Writing](#))
 - Microsoft Data Analysis (SQL, Excel, Power BI):** Microsoft (ID: [VXYOBGDSIQOO](#))
 - Google AI Professional Certificate:** Google (ID: [FW6A17MGN8U3](#))
 - Getting Started with Data Analytics on AWS:** AWS (ID: [OR8IQB4CTI1](#))
 - Image Classification on AWS AutoGluon:** Coursera (ID: [MVVF8ARA6H2C](#))
- Enhanced DBS Certification:** Active & Verified ([All UK background checks cleared](#))
 - Gemini for Developers Specialization:** Google DeepMind (ID: [DMVKPELEY3J2](#))
 - AI for Mechanical Engineers:** Univ. of Michigan (ID: [BUUPCEOS9DG1](#))
 - Scikit-Learn for ML Classification Problems:** Coursera (ID: [8KLLIRRLXTWN](#))
 - Data Engineering Master Blueprints:** GitHub ([8 Master Architectural Guides](#))

Languages: English (Fluent), Hindi (Native), Bengali (Native)

GitHub Portfolios: [github.com/Xclipxz07](#) • References available upon request